



Additional Practice Questions

Capacitance

Important note: *denotes questions that may require knowledge from later chapters in order to solve them.

I. MCQs

1. A capacitor of capacitance C_1 is charged to a potential difference of 100 V and then disconnected. When a second uncharged capacitor of capacitance C_2 is connected in parallel across it, the new p.d. is 60 V.

What is the value of the ratio $\frac{C_1}{C_2}$?

- A $\frac{3}{5}$ B $\frac{2}{3}$ C $\frac{3}{2}$ D $\frac{5}{3}$

2. Two capacitors of capacitance C and $2C$, are charged to potential differences V and $2V$ respectively. If the two positive plates are connected together and the two negative plates are connected together, then this system of capacitors

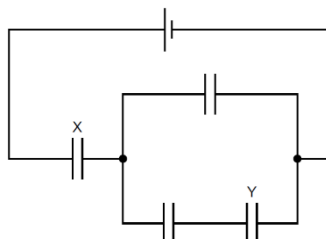
- A gains charge but loses energy
B gains energy but loses charge
C loses charge but energy remains constant
D loses energy but charge remains constant

3. A capacitor of capacitance $160 \mu\text{F}$ is charged to a potential difference of 200 V and then connected across a discharge tube, which conducts until the potential difference across it has fallen to 100 V.

The energy dissipated in the tube is

- A 0.80 J B 2.4 J C 3.2 J D 4.8 J

4. Four identical capacitors are connected to a cell as shown.



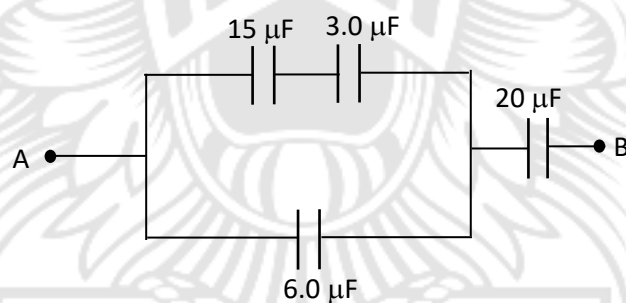
The charge stored on capacitor X is $6\ \mu\text{C}$.

What is the charge stored on capacitor Y?

- A $2\ \mu\text{C}$ B $3\ \mu\text{C}$ C $4\ \mu\text{C}$ D $6\ \mu\text{C}$

II. Structured Questions

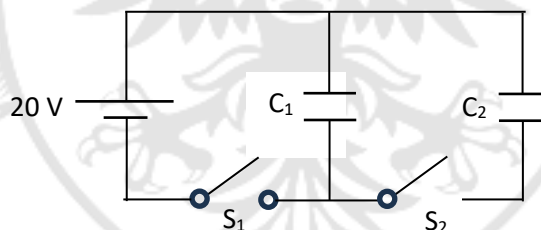
1. Four capacitors are connected as shown.



- a. Determine the equivalent capacitance C_{AB} between points A and B. [2]
 b. The potential difference V_{AB} across A and B is $15.0\ \text{V}$. Calculate the charge on each capacitor. [3]

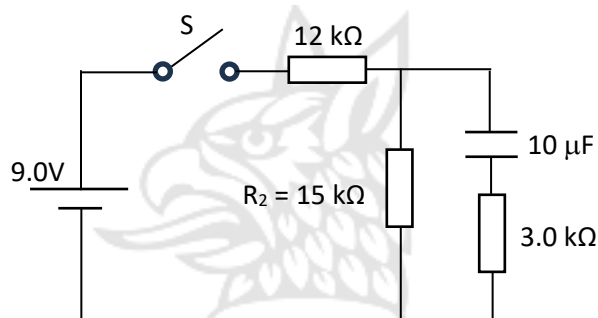
[Total: 5]

2. In the circuit shown, capacitor C_1 of capacitance $6.0\ \mu\text{F}$ is first charged by closing switch S_1 . Switch S_1 is then opened, and the charged capacitor is connected to the uncharged capacitor C_2 , of capacitance $3.0\ \mu\text{F}$ by closing switch S_2 . Calculate the initial charge stored in capacitor C_1 and the final charge on each capacitor.



[4]

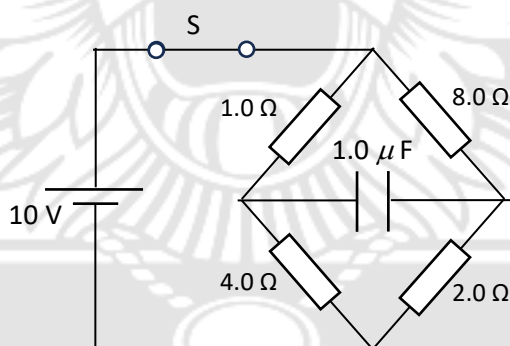
3. In the circuit shown, the switch S was closed for a sufficiently long time for the capacitor to be fully charged.



- a. Determine the charge Q on the capacitor. [2]
- b. The switch S is then opened at time $t = 0$.
 - i. Write down an equation for the current I_{R_2} in resistor R_2 as a function of time t . [3]
 - ii Calculate the time taken for the charge on the capacitor to fall to 20% of its initial value. [2]

[Total: 7]

4. The circuit shown below was left closed for a long time.



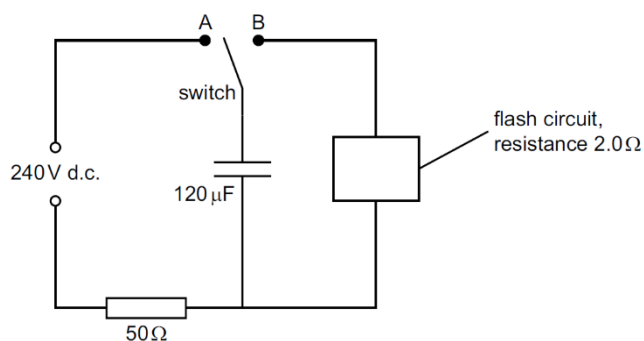
- a. What is the potential difference across the capacitor? [2]
- b. The switch is then opened. Determine the time taken for the p.d. across the capacitor to decrease to 10% of its initial value. [2]

[Total: 4]

5 a. Define capacitance.

[1]

b. A capacitor is connected in a circuit used for a flash camera as shown below.



- i. When the switch is connected to terminal A, the capacitor charges. The switch automatically connects to terminal B when the charge in the capacitor reaches 95% of its maximum possible charge at this voltage. Calculate the time taken for the capacitor to reach 95% of its maximum charge. [2]
- ii. The capacitor now discharges through the flash circuit. Calculate the maximum amount of energy that the capacitor can deliver to the flash circuit. [2]

[Total: 5]